

Data Analytics V.IMP Questions

EduShine Classes – Arman Ali

Unit I: Introduction to Data Analytics & Lifecycle

1. Explain the phases of the Data Analytics Lifecycle in detail.
2. Classify the types of data (structured, semi-structured, unstructured) with suitable examples.
3. Discuss the characteristics and need of Big Data in today's world.
4. What are the key roles required for a successful data analytics project?
5. Write a short note on modern data analytics tools and their applications.
6. Compare and contrast traditional analytics structure to modern analytics architecture.
7. Explain various phases of Data Analytics Life Cycle.
8. Explain multivariate analysis and Bayesian network.
9. How RDBS is different from NoSQL?
10. Explain Datar-Gionis-Indyk-Motwani (DGIM) algorithm for counting ones in a window.

Unit – 2 Data Analysis

1. Differentiate between regression and classification models with suitable examples.
2. Explain Bayesian modeling and its applications in data analytics.
3. What is Principal Component Analysis (PCA)? Explain with an example.
4. Discuss the working and advantages of neural networks in data analysis.
5. Write a note on fuzzy logic and fuzzy decision trees with example.
6. Explain Bernoulli sampling with its algorithm.

7. What are the different components of a general stream processing model? List few sources of streaming data.
8. What is Prediction error? With the help of suitable example explain prediction error in classification and regression.
9. Differentiate between Crisp logic and Fuzzy logic.
10. What are the different kernel methods of Data Analytics?

Unit – 3 Mining Data Stream

1. What is a data stream? Explain stream data model and its architecture.
2. Explain different stream mining tasks with suitable examples.
3. What is Real-Time Analytics Platform (RTAP)? Describe its working.
4. Describe time series analysis and its importance in data analytics.
5. Explain techniques for estimating moments and detecting anomalies in streaming data.
6. Why PCY algorithm is preferred over Apriori algorithm?
7. Explain SON algorithm to find all or most frequent item sets using at most two passes.
8. Explain Apache Hadoop, KNIME & Open Refine in detail.
9. Explain architecture of Google File System.
10. List out Big Data technologies.

Unit – 4 Frequent Item Set & Clustering

1. Explain Apriori Algorithm with example. How is it used in association rule mining?
2. Describe K-means clustering algorithm with step-by-step working.
3. Differentiate between K-means and hierarchical clustering.
4. Explain FP-Growth algorithm and how it overcomes Aprioris limitations.
5. Write a note on pattern-based clustering and clustering in non-Euclidean space.
6. In the context of stream data, explain different approaches for counting distinct elements. How do these methods address challenges associated with continuously changing data?
7. Describe the concept of counting uniqueness in a window in the context of stream processing. How does this relate to measuring the frequency and uniqueness of elements within a specified time frame?
8. What is Prediction Error? With the help of suitable examples, explain prediction error in classification and regression.
9. Explain SON algorithm to find all or most frequent item sets using at most two passes.

Unit – 5 Frame Work & Visualisation & Introduction to R

1. Compare Hadoop and Spark in terms of architecture and data processing.
2. What is MapReduce? Explain its working with a simple example.
3. Explain the role of HDFS and NoSQL in big data analytics.
4. Why is data visualization important? Discuss types of visualization with examples.
5. Write a short note on any three visualization tools used in data analytics

6. Compare the strengths and weaknesses of hierarchical clustering and K-means clustering. Under what circumstances would one technique be preferred over the other, and why?
7. Provide an in-depth comparison between the CLIQUE and ProCLUS clustering algorithms. How do these methods handle challenges such as noise, outliers, and varying cluster shapes?
8. Explore the challenges and considerations when performing clustering in non-Euclidean spaces. How do distance metrics and similarity measures differ in non-Euclidean environments, and what impact does this have on clustering outcomes?
9. How do interactive techniques contribute to the exploration and analysis of large datasets? Provide examples of systems or tools that leverage interactive approaches effectively.
10. Draw and discuss the architecture of Hive in detail.
11. What are the approaches to integrate the human in data exploration process to realize different types of approaches to visual data mining?

BY- ARMAN ALI

B.Tech.
(SEM V) THEORY EXAMINATION 2022-23
DATA ANALYTICS

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**2 x 10 = 20**

- (a) What are the common open source tools for the model building phase?
- (b) What is decision tree?
- (c) What is learning rate?
- (d) What is rule induction?
- (e) How can you deal with uncertainty?
- (f) What is DSMS?
- (g) Write names of two visualization tools.
- (h) Explain the principle behind Hierarchical clustering technique.
- (i) Differentiate between Pig and SQL.
- (j) Write a short note on R graphical user interface.

SECTION B

2. Attempt any three of the following:**10x3 = 30**

- (a) Compare and contrast traditional analytics structure to modern analytics architecture.
- (b) Explain multivariate analysis and Bayesian network.
- (c) Explain Datar-Gionis-Indyk-Motwani (DGIM) algorithm for counting oneness in a window.
- (d) Why PCY algorithm is preferred over Apriori algorithm?
- (e) How RDBS is different from NoSQL?

SECTION C

3. Attempt any one part of the following:**10x1 = 10**

- (a) Explain Apache Hadoop , KNIME & Open refine in detail.
- (b) Explain various phases of Data Analytics Life Cycle.

4. Attempt any one part of the following:**10x1 = 10**

- (a) Differentiate between Crisp logic and Fuzzy logic.
- (b) What are the different kernel methods of Data Analytics?

5. Attempt any *one* part of the following:

10x1 = 10

- (a) Explain Bernoulli sampling with its algorithm.
- (b) What are the different components of a general stream processing model? List few sources of streaming data .

6. Attempt any *one* part of the following:

10x1 = 10

- (a) What is Prediction error ? With the help of suitable example explain prediction error in classification and regression.
- (b) Explain SON algorithm to find all or most frequent item sets using at most two passes.

7. Attempt any *one* part of the following:

10x1 = 10

- (a) Draw and discuss the architecture of Hive in detail.
- (b) What are the approaches to integrate the human in data exploration process to realize different types of approaches to visual data mining?



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BTECH
(SEM V) THEORY EXAMINATION 2023-24
DATA ANALYTICS

TIME: 3 HRS**M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**2 x 10 = 20**

Q no.	Question	Marks
a.	How have advancements in technology contributed to the scalability of analytics?	2
b.	What are the sources of data in the context of data analytics?	2
c.	Elaborate on the mathematical foundations of support vector machines.	2
d.	Discuss the advantages of using Bayesian methods in real-world applications.	2
e.	Elaborate on the methods used for filtering streams in real-time analytics.	2
f.	What considerations should be taken into account when implementing sampling techniques for stream data?	2
g.	How do stream-based algorithms differ from batch processing algorithms in the context of frequent itemset mining?	2
h.	What are the challenges associated with implementing Apriori in scenarios with limited available memory?	2
i.	Explain the role of Hive in the Hadoop ecosystem.	2
j.	How does the MapReduce framework facilitate distributed processing?	2

SECTION B

2. Attempt any three of the following:**10 x 3 = 30**

a.	Describe the characteristics of data that are relevant in the field of data analytics. How do these characteristics impact the analysis process?	10
b.	Explain the concept of Bayesian networks and their applications in modeling probabilistic relationships among variables. Discuss how Bayesian networks can be constructed from data and used for reasoning under uncertainty.	10
c.	Provide an overview of Real-time Analytics Platforms applications, emphasizing their role in processing continuous data streams. How do these platforms support the development of real-time analytics solutions?	10
d.	Compare the strengths and weaknesses of hierarchical clustering and K-means clustering. Under what circumstances would one technique be preferred over the other, and why?	10
e.	Discuss the concept of sharding in the context of NoSQL databases. How does sharding contribute to scalability, and what challenges does it address?	10

SECTION C

3. Attempt any one part of the following:**10 x 1 = 10**

a.	Explain the concept of generalization in neural networks. How does it relate to the trade-off between bias and variance, and what strategies can be employed to enhance generalization performance?	10
b.	Provide a detailed explanation of how fuzzy logic is used to extract models from data. Discuss the advantages of fuzzy modeling in capturing uncertainty and handling imprecise information in comparison to traditional crisp models.	10



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BTECH
(SEM V) THEORY EXAMINATION 2023-24
DATA ANALYTICS

TIME: 3 HRS**M.MARKS: 100****4. Attempt any one part of the following: 10 x 1 = 10**

a.	In the context of stream data, explain different approaches for counting distinct elements. How do these methods address challenges associated with continuously changing data?	10
b.	Describe the concept of counting uniqueness in a window in the context of stream processing. How does this relate to measuring the frequency and uniqueness of elements within a specified time frame?	10

5. Attempt any one part of the following: 10 x 1 = 10

a.	Provide an in-depth comparison between the CLIQUE and ProCLUS clustering algorithms. How do these methods handle challenges such as noise, outliers, and varying cluster shapes?	10
b.	Explore the challenges and considerations when performing clustering in non-Euclidean spaces. How do distance metrics and similarity measures differ in non-Euclidean environments, and what impact does this have on clustering outcomes?	10

6. Attempt any one part of the following: 10 x 1 = 10

a.	How do interactive techniques contribute to the exploration and analysis of large datasets? Provide examples of systems or tools that leverage interactive approaches effectively.	10
b.	Discuss the role of NoSQL databases in handling unstructured data. Provide examples of scenarios where NoSQL databases outperform traditional relational databases.	10

7. Attempt any one part of the following: 10 x 1 = 10

a.	Differentiate between analysis and reporting in the context of data analytics. How do these two aspects contribute to the overall understanding of data?	10
b.	Explore modern data analytic tools and their functionalities. How have these tools transformed the landscape of data analytics?	10